

## Supplementary information

### Disease-blind reconstruction distinguishes reproducible interferon remodeling from unstable B-cell state assignments in systemic lupus erythematosus

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#### Supplementary overview

This single Supplementary Information file contains Tables S1-S9 and Figures S1-S10 supporting the main Article. All analytical methods required to interpret the work are reported in the main manuscript. Machine-readable figure source data, regulator sensitivity results and complete statistical outputs are supplied separately as Supplementary Data 1-3. Original external-mapping sensitivity outcomes are superseded, and no corrected disease effect was estimated after calibration failed.

#### Supplementary Table S1 | Dataset roles and inferential units

| Resource           | Role                              | Biological unit                            | Active scope                                                                       |
|--------------------|-----------------------------------|--------------------------------------------|------------------------------------------------------------------------------------|
| GSE174188          | Discovery and internal validation | Sample-cohort stratum; donor sensitivities | Disease-blind B_CONV/B_ASC analysis scaffold, composition and B_CONV transcription |
| GSE135779          | Independent SLE validation        | Donor                                      | Broad conventional-B analogue; childhood primary                                   |
| CollecTRI/OmniPath | Curated regulator prior           | Ranked genes within contrast               | Prespecified eight-regulator, three-contrast family                                |
| MSigDB M5911       | Orthogonal response prior         | Ranked genes within contrast               | Hallmark interferon-alpha response enrichment                                      |
| GSE23307           | Orthogonal perturbation support   | Paired profile within donor                | Descriptive IFN-beta response, n=2 donors                                          |

#### Supplementary Table S2 | Claim boundaries

| Supported                                                                         | Not supported                                                                                    |
|-----------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------|
| Disease-blind B_CONV/B_ASC analysis scaffold with quantified boundary sensitivity | Universally reproducible B_CONV/B_ASC taxonomy or stable hard naive, memory or atypical subtypes |
| Primary B_ASC contrast lacks statistical support                                  | General B_ASC expansion in SLE                                                                   |
| Replicated IFN/ISG remodeling within B_CONV                                       | Genome-wide shared disease transcriptome                                                         |
| ULM STAT1/STAT2 activity concordant across three contrasts                        | Causal TF activation or direct binding                                                           |
| IFN-centred response evidence                                                     | A unique initiating IFN ligand in SLE                                                            |

**Supplementary Table S3 | Quantitative anchors and prespecified boundaries**

| Analysis                              | Result                                                                                                                                    |
|---------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| Frozen-representation identity policy | minimum mapped ARI 0.990; minimum agreement 0.9998; minimum state-median Jaccard 0.991                                                    |
| End-to-end identity sensitivity       | prespecified criterion not met; minimum mapped ARI 0.930; minimum agreement 0.9988; B_ASC median Jaccard 0.930 below 0.95 criterion       |
| Boundary propagation                  | primary B_ASC odds-ratio range 0.896-0.967, all intervals include one; primary B_CONV IFN/ISG range 0.836-0.845, all intervals above zero |
| Primary B_ASC composition             | 43 controls and 47 source-defined managed SLE; odds ratio 0.947; 95% CI 0.636-1.410; P=0.787                                              |
| GSE174188 primary IFN/ISG             | effect 0.837; 95% CI 0.525-1.148; $q=2.98 \times 10^{-6}$                                                                                 |
| GSE174188 donor-nonoverlap IFN/ISG    | effect 1.086; $q=3.61 \times 10^{-4}$                                                                                                     |
| GSE135779 childhood IFN/ISG           | effect 1.042; 95% CI 0.681-1.402; $q=2.98 \times 10^{-6}$                                                                                 |
| Cross-dataset genome-wide agreement   | 4,410 genes; Spearman rho=0.026                                                                                                           |
| M5911 enrichment                      | NES 3.187, 3.050 and 3.527                                                                                                                |
| GSE23307 perturbation                 | donor effects 3.294 and 3.666; 12/12 genes positive in each                                                                               |

**Supplementary Table S4 | Correlation-aware core-regulator sensitivity**

| Contrast                   | Regulator | Matched targets | Inter-gene correlation | CAMERA BH q | FRY BH q  |
|----------------------------|-----------|-----------------|------------------------|-------------|-----------|
| GSE174188 primary          | STAT1     | 98              | 0.0362                 | 0.0263      | 6.043e-06 |
| GSE174188 primary          | STAT2     | 14              | 0.1225                 | 0.1355      | 4.909e-05 |
| GSE174188 donor-nonoverlap | STAT1     | 129             | 0.0209                 | 0.0263      | 8.49e-06  |
| GSE174188 donor-nonoverlap | STAT2     | 19              | 0.1191                 | 0.0263      | 4.277e-07 |
| GSE135779 childhood        | STAT1     | 161             | 0.0301                 | 0.0263      | 2.685e-05 |
| GSE135779 childhood        | STAT2     | 20              | 0.1223                 | 0.03026     | 1.231e-05 |

CAMERA and FRY directions were positive in all six tests. CAMERA passed BH correction in five of six; the explicit exception was GSE174188 primary STAT2 ( $q=0.1355$ ). FRY passed BH correction in all six.

**Supplementary Table S4B | IFN-overlap-depletion summary**

| Depletion                  | Method | Positive direction | Dedicated $q<0.05$ | Main qualification                                                     |
|----------------------------|--------|--------------------|--------------------|------------------------------------------------------------------------|
| Frozen 12-gene IFN/ISG arm | ULM    | 6/6                | 6/6                | Minimum slope retention 53.5%; all 95% confidence intervals above zero |
| Frozen 12-gene IFN/ISG arm | CAMERA | 6/6                | 5/6                | Discovery STAT2 $q=0.326$                                              |

| Depletion                  | Method | Positive direction | Dedicated $q < 0.05$ | Main qualification                                                                      |
|----------------------------|--------|--------------------|----------------------|-----------------------------------------------------------------------------------------|
| Frozen 12-gene IFN/ISG arm | FRY    | 6/6                | 6/6                  | Direction and corrected support retained                                                |
| M5911                      | ULM    | 6/6                | 5/6                  | Discovery STAT2 retained 8/14 targets; slope 0.391, 95% CI -0.745 to 1.526; $q = 0.500$ |
| M5911                      | CAMERA | 6/6                | 2/6                  | Broad attenuation across contrasts; discovery STAT2 $q = 0.623$                         |
| M5911                      | FRY    | 6/6                | 5/6                  | Discovery STAT2 $q = 0.099$                                                             |

All 11 depleted ULM models retaining at least ten targets preserved direction in every leave-one-target analysis. These sensitivities support robustness beyond the narrow 12-gene arm but not independence from the broader interferon-response transcriptome.

### Supplementary Table S5 | Main-figure source-data map

| Figure                   | Evidence basis                                                       | Machine-readable source                  |
|--------------------------|----------------------------------------------------------------------|------------------------------------------|
| Figure 1                 | Disease-blind identity stability and two-compartment adjudication    | Figure1_source_data.csv                  |
| Figure 2                 | Sample-level composition and asserted 43/47 primary groups           | Figure2_source_data.csv                  |
| Figure 3                 | Raw-count pseudobulk transcription with explicit tested-gene symbols | Figure3_source_data.csv                  |
| Figure 4                 | Independent GSE135779 validation and influence analyses              | Figure4_source_data.csv                  |
| Figure 5                 | Regulatory and orthogonal response evidence                          | Figure5_source_data.csv                  |
| Supplementary Figure S8  | STAT1/STAT2 IFN-overlap-depletion sensitivity                        | Supplementary_Figure_S8_source_data.csv  |
| Supplementary Figure S9  | End-to-end identity boundary and downstream propagation              | Supplementary_Figure_S9_source_data.csv  |
| Supplementary Figure S10 | Corrected reference calibration and unresolved external transfer     | Supplementary_Figure_S10_source_data.csv |

### Supplementary Table S6 | Reproducibility record

| Component                          | Reader-accessible record                                                          |
|------------------------------------|-----------------------------------------------------------------------------------|
| Main and supplementary figure data | Supplementary Data 1 with SHA-256 manifest                                        |
| Complete statistical results       | Supplementary Data 3 with 12 gene-level branches and 12 sanitized design matrices |
| Correlation-aware sensitivity      | Supplementary Data 2                                                              |

| Component                                    | Reader-accessible record                                                                                                                                              |
|----------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| End-to-end identity and boundary propagation | Supplementary Data 3, identity_robustness/                                                                                                                            |
| Analysis code and decisions                  | <a href="https://github.com/1209433622cz-maker/sle-bcell-remodeling">https://github.com/1209433622cz-maker/sle-bcell-remodeling</a>                                   |
| Environment reconstruction                   | Pinned scientific and release environments documented in REPRODUCIBILITY.md                                                                                           |
| Version-specific archive                     | Zenodo <a href="https://doi.org/10.5281/zenodo.22151739">https://doi.org/10.5281/zenodo.22151739</a> ; matches the frozen manuscript, figures and statistical outputs |

## Supplementary Table S7 | Statistical tests and multiplicity families

| Result family                       | Biological unit                              | Test or estimator                                     | Sidedness                                    | Multiplicity                                      | Role                               |
|-------------------------------------|----------------------------------------------|-------------------------------------------------------|----------------------------------------------|---------------------------------------------------|------------------------------------|
| B_ASC composition                   | Sample-cohort stratum                        | Beta-binomial Wald                                    | Two-sided                                    | BH across three frozen base contrasts             | Primary composition inference      |
| Gene-level expression               | Donor/sample pseudobulk                      | edgeR robust quasi-likelihood F                       | Two-sided                                    | BH within tested genes per contrast               | Gene-level inference               |
| Four frozen programs                | Donor/sample pseudobulk                      | OLS with HC3                                          | Two-sided                                    | BH across four programs per analysis              | Program inference                  |
| TF-target activity                  | Ranked gene statistics                       | Signed-target slope t test                            | Two-sided                                    | Global BH across 24 tests                         | Confirmatory regulatory evidence   |
| STAT1/STAT2 sensitivity             | Ranked gene statistics                       | CAMERA and FRY                                        | Positive-direction                           | Separate BH families of six per method            | Correlation-aware sensitivity      |
| STAT1/STAT2 overlap depletion       | Ranked gene statistics and pseudobulk counts | ULM, CAMERA and FRY                                   | Two-sided ULM; positive-direction CAMERA/FRY | Separate BH family of six per branch and method   | Post-freeze sensitivity            |
| End-to-end identity reproducibility | Cell resample within technical library       | ARI, AMI, mapping agreement, state Jaccard and recall | Not tested                                   | Five unchanged threshold criteria                 | Disease-blind robustness boundary  |
| Broad-state boundary propagation    | Sample-cohort stratum                        | Frozen beta-binomial and TMM logCPM/HC3 models        | Two-sided intervals                          | No new multiplicity family; same-data sensitivity | Assignment-uncertainty sensitivity |
| M5911                               | Ranked gene statistics                       | 10,000-permutation preranked test                     | Positive-direction                           | Descriptive BH across three contrasts             | Orthogonal support                 |
| GSE23307                            | Paired donor                                 | Mean paired $\log_2(x+1)$ effect                      | Not tested                                   | None; n=2                                         | Descriptive perturbation           |

**Supplementary Table S8 | Full statistical-results archive map**

| Directory                    | Contents                                                                                                                                                             | Files |
|------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| gene_level_results           | Complete filterByExpr gene results for GSE174188 and GSE135779 branches                                                                                              | 12    |
| sanitized_design_matrices    | Direct-identifier-free analysis designs                                                                                                                              | 12    |
| composition                  | Frozen coefficients, contrasts, predictions, sensitivities, leave-one-out and diagnostics                                                                            | 8     |
| transcription                | Program, ranked-list, influence and concordance results                                                                                                              | 18    |
| regulatory_and_orthogonal    | ULM, target influence, CAMERA/FRY, M5911 and GSE23307 summaries                                                                                                      | 9     |
| statistical_framework        | Machine-readable test and multiplicity map                                                                                                                           | 1     |
| identity_robustness          | Aggregate and replicate-level end-to-end identity metrics, boundary propagation results and integrity records                                                        | 101   |
| external_mapping_calibration | Corrected feature and parameter tables, full confidence-calibration family, aggregate selection diagnostics, provenance, arithmetic recount and publication boundary | 20    |

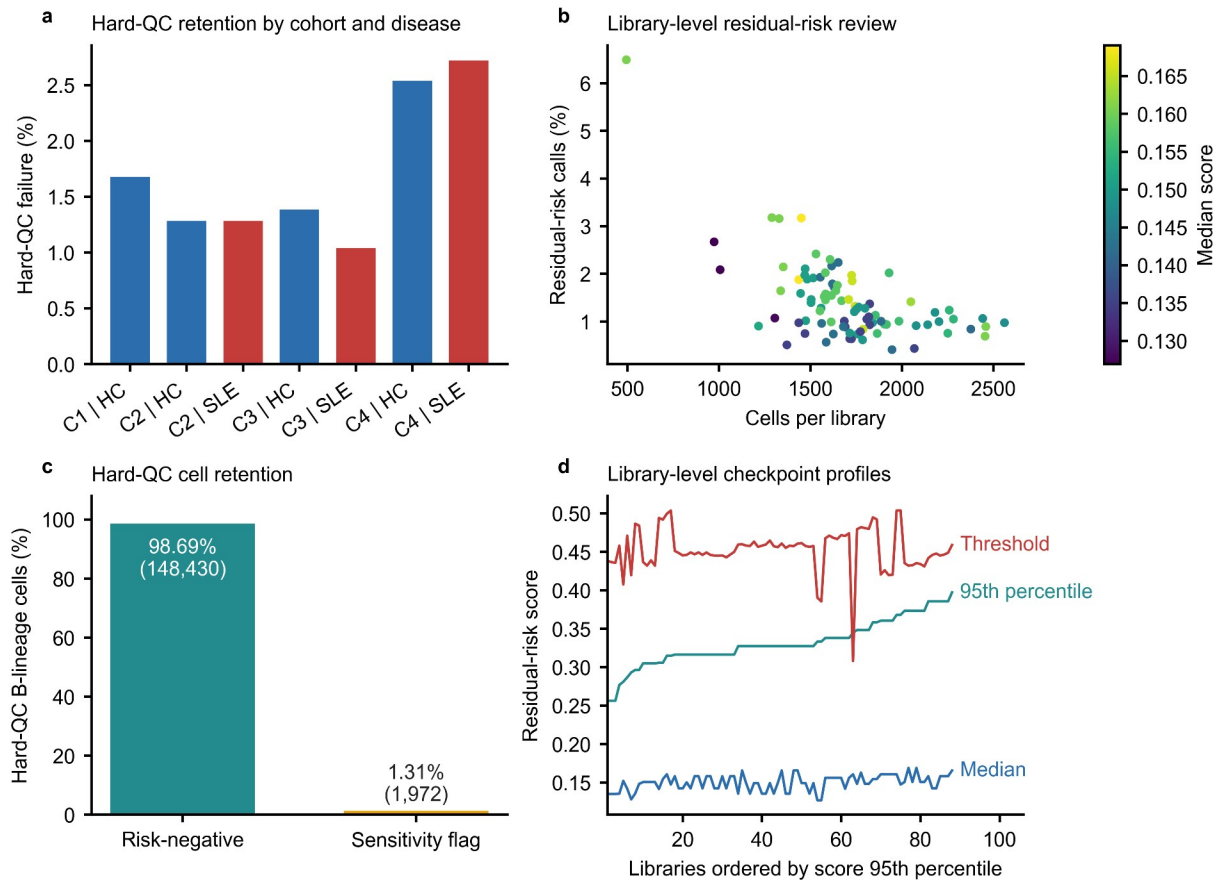
**Supplementary Table S9 | Reference-calibrated external mapping boundary**

| Component                             | Corrected diagnostic                                                                              | Interpretation                                                                          |
|---------------------------------------|---------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------|
| External input                        | 56 matrices; 363,083 cells                                                                        | Complete source matrix coverage                                                         |
| External QC and selection             | 353,527 QC-passing cells; 36,630 B-lineage candidates                                             | No source labels or disease fields parsed                                               |
| Reference training                    | 13,000 B_CONV and 1,300 B_ASC; 258 donors; 601 features                                           | Both algorithms share reference and features                                            |
| Normalization                         | Full-library log1p(CP10K) before feature subsetting in both datasets                              | Corrects reference feature-only denominator                                             |
| Elastic-net calibration               | Diagnostic threshold 0.95; coverage 0.941958; B_CONV precision 0.996450; B_ASC precision 0.885210 | No eligible candidate: state precision must be at least 0.90 and coverage at least 0.80 |
| Centroid calibration                  | Margin 0.117767; coverage 0.95; B_CONV precision 0.992374; B_ASC precision 1.0                    | Eligible, but cannot replace required primary mapper                                    |
| Mean reference-fold balanced accuracy | Elastic net 0.960569; centroid 0.950293                                                           | Calibration diagnostics; not independent performance estimates                          |
| Outcome policy                        | Corrected disease outcomes not estimated                                                          | No corrected effect, confidence interval, P or q value exists                           |
| Prior outcome exposure                | Original sensitivity outcomes were known before correction                                        | Post-outcome correction; not prospective validation                                     |
| Evidence boundary                     | Primary GSE135779 replication remains source-label-defined                                        | Superseded uncorrected outcomes excluded from supporting evidence                       |

Full candidate calibration, donor-grouped cross-validation, reference normalization audit and corrected input/prediction hashes are retained with the analysis code. The original source-label-based pseudobulk effect is a standardized HC3 model estimate; the unexecuted sensitivity would instead compare donor mean cell-level  $\log_{10}(\text{CP10K})$  scores. They are different estimands and must not be compared as effect attenuation. No new multiplicity family was evaluated after the calibration HOLD.

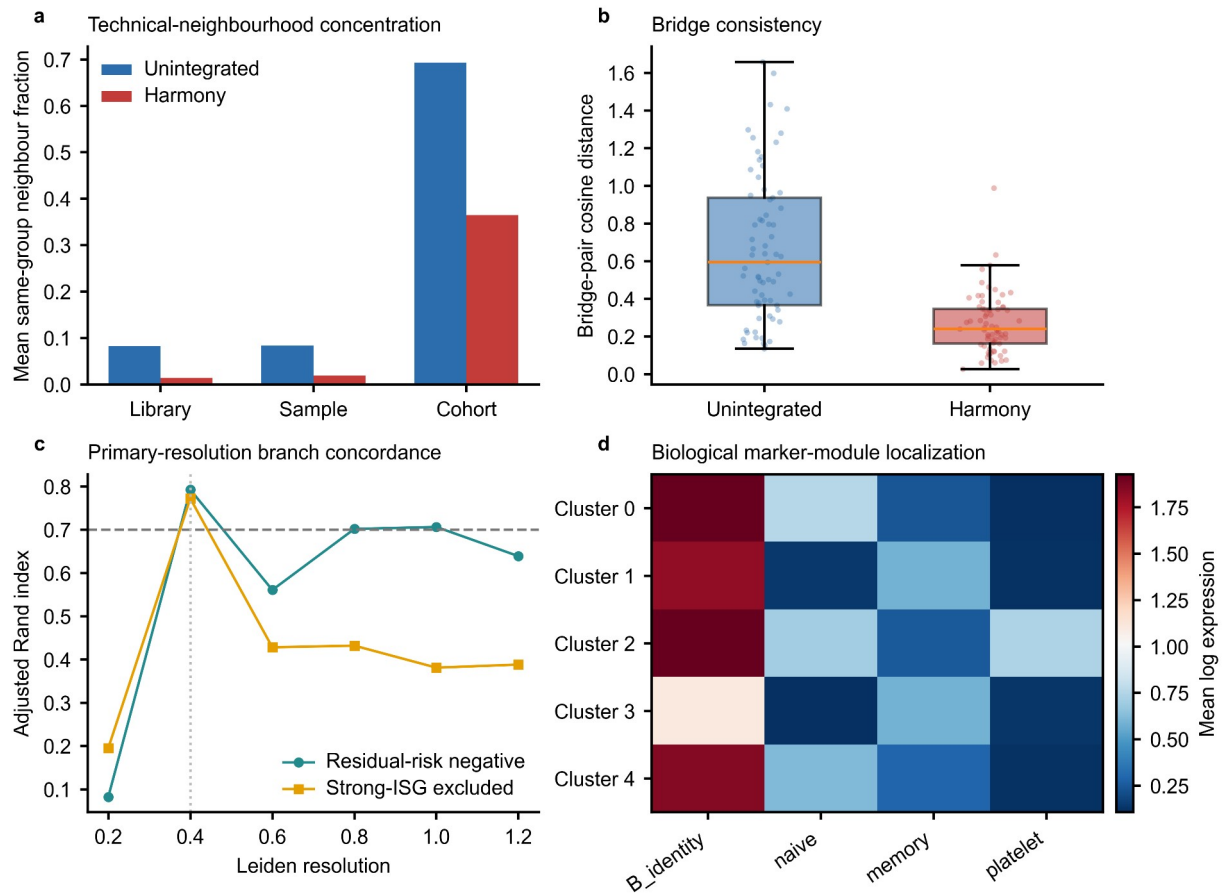
## Supplementary Figure S1 | Source integrity and hard-quality-control diagnostics

**a**, Hard-QC failure fractions by processing cohort and disease group. **b**, Residual-risk call fraction versus cells per library, coloured by the library median score. **c**, Percent and exact number of retained risk-negative cells and sensitivity-only residual-risk calls among 150,402 hard-QC B-lineage cells. **d**, Median, 95th-percentile and threshold residual-risk scores for all 88 libraries.



## Supplementary Figure S2 | Representation and bridge diagnostics

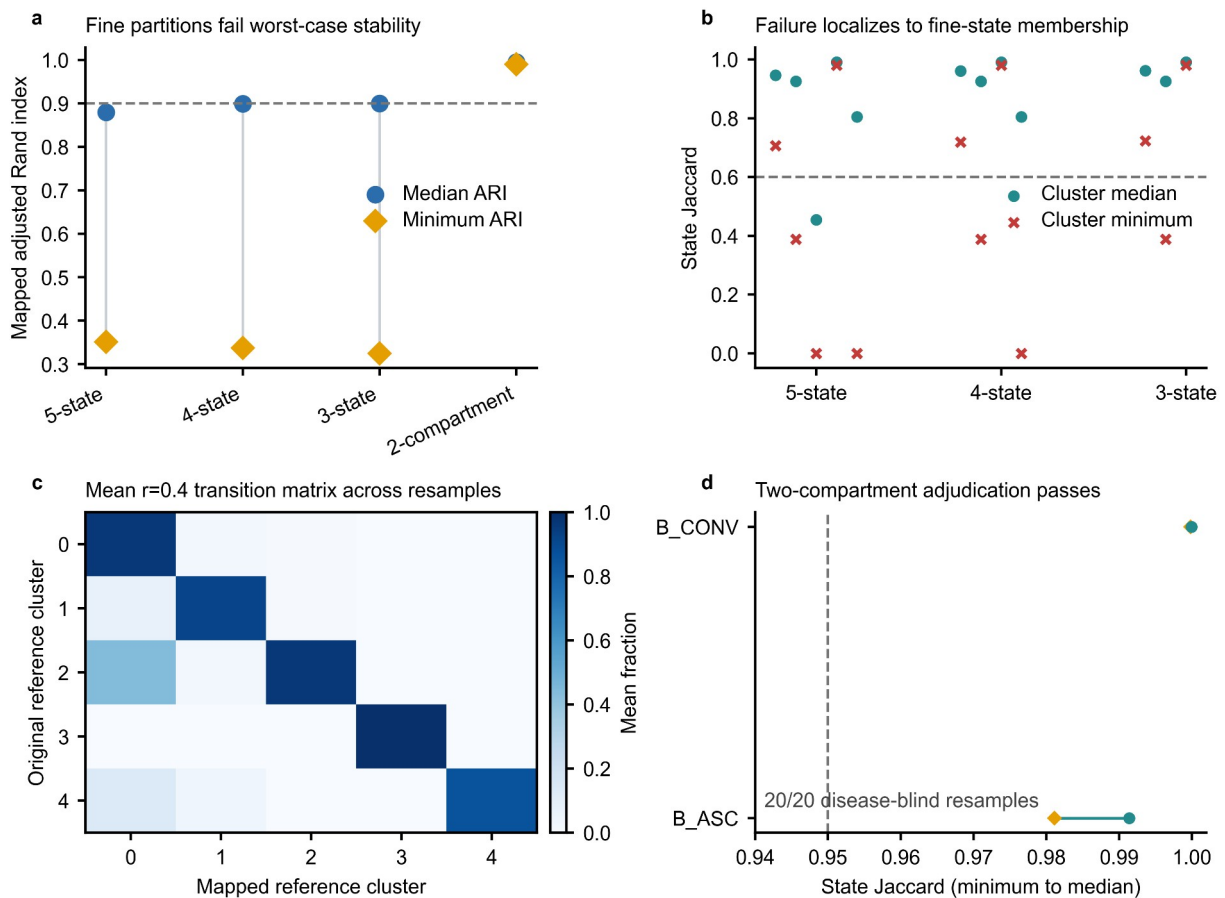
**a**, Same-group neighbourhood concentration before and after Harmony adjustment. **b**, Cross-cohort bridge-pair cosine-distance distributions. **c**, Disease-blind branch concordance across Leiden resolutions for residual-risk-negative and strong-ISG-excluded branches; the vertical guide marks the primary resolution. **d**, Biological marker-module localization across the five fine clusters considered before stability adjudication.





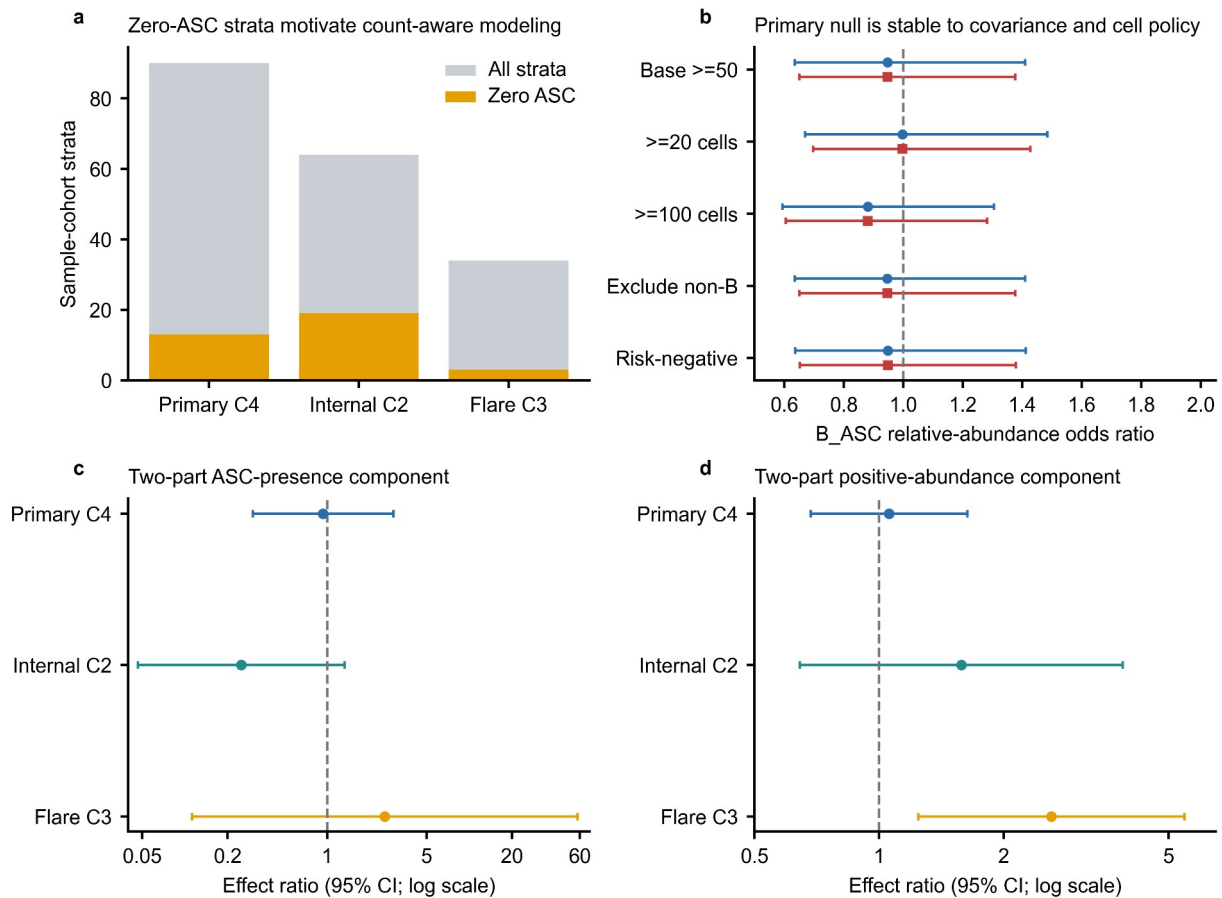
## Supplementary Figure S3 | Disease-blind identity adjudication

**a**, Median and worst-case mapped ARI for five-, four-, three- and two-level policies across frozen-representation resamples. **b**, Median and minimum cluster Jaccard values show localization of failure to fine-state membership. **c**, Mean transition matrix from original resolution-0.4 clusters to mapped reference clusters. **d**, Minimum-to-median Jaccard intervals for the B\_CONV and B\_ASC analysis states across 20 frozen-representation resamples. Highly variable genes, principal components and Harmony coordinates were not recomputed in this figure.



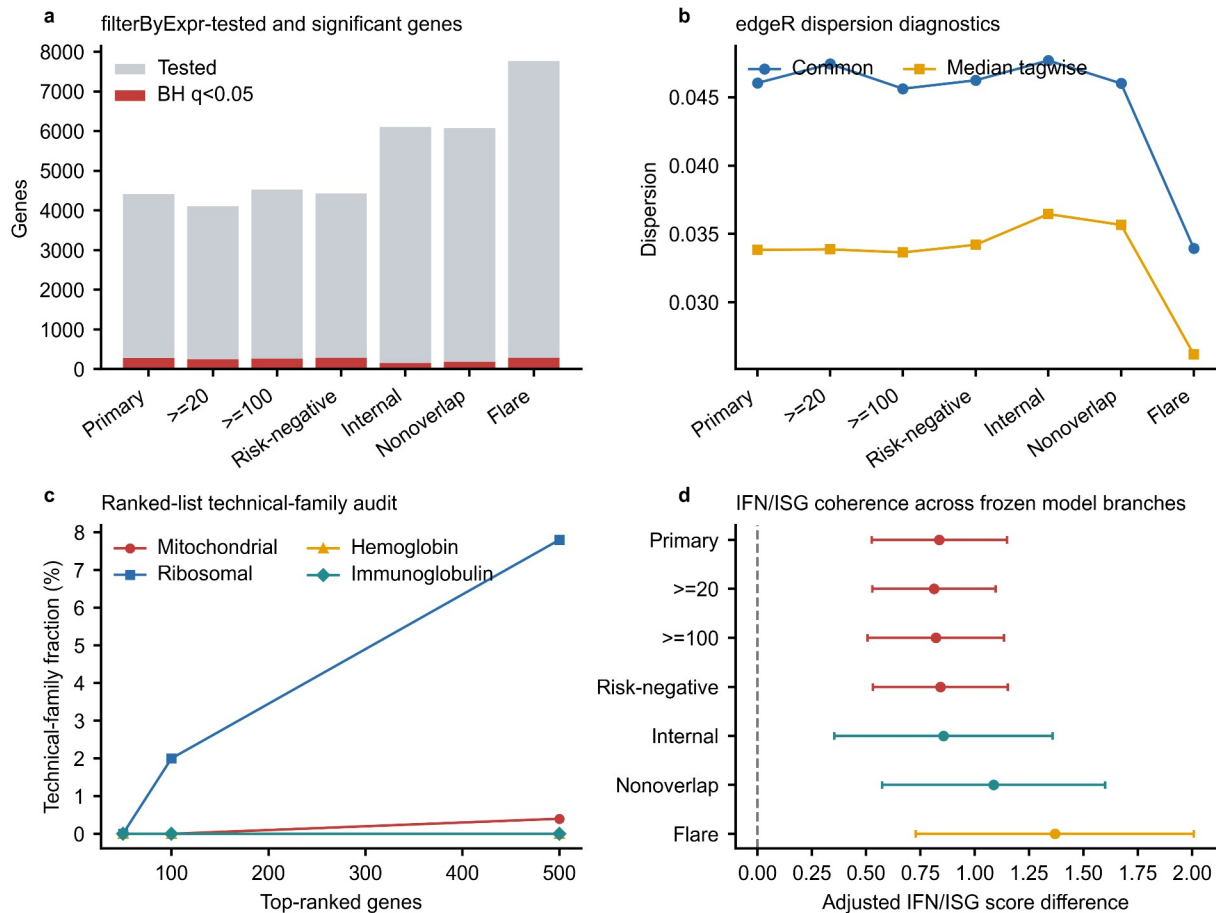
## Supplementary Figure S4 | Composition-model diagnostics

**a**, Total and zero-ASC sample-cohort strata in the three base contrasts. **b**, Primary B\_ASC odds ratios under support, explicit non-B and residual-risk policies using observed-information and HC1 covariance. **c**, Firth-logistic ASC-presence sensitivity. **d**, Positive-only abundance sensitivity using HC3 uncertainty. Panels c-d use logarithmic ratio axes with the null fixed at one; the two-part models are sensitivity analyses and do not replace the frozen beta-binomial primary model.



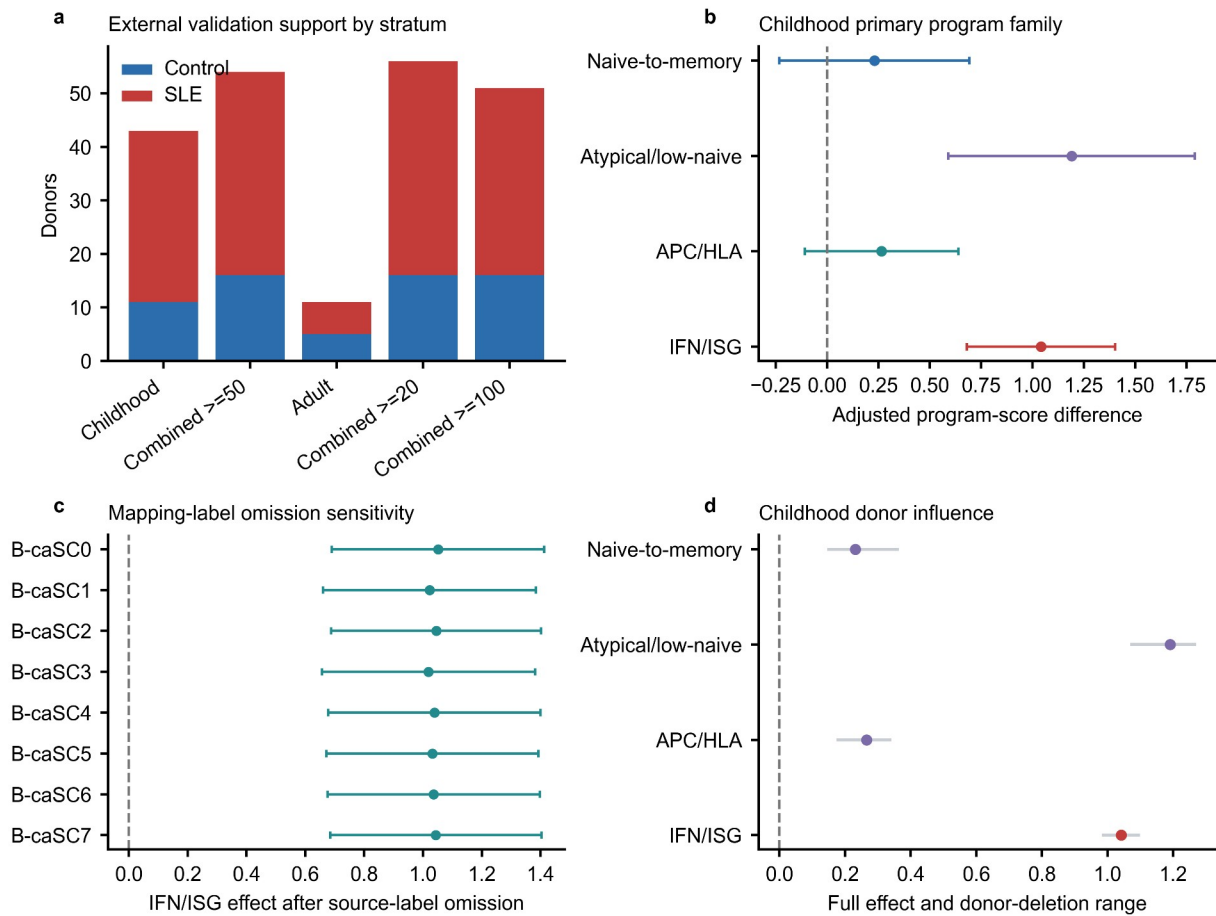
## Supplementary Figure S5 | Pseudobulk and ranked-list diagnostics

**a**, Numbers of filterByExpr-tested and BH-significant genes across seven GSE174188 branches. **b**, Common and median tagwise edgeR dispersions. **c**, Mitochondrial, ribosomal, haemoglobin and immunoglobulin fractions among increasingly long primary ranked lists. **d**, IFN/ISG effects and 95% confidence intervals across frozen branches.



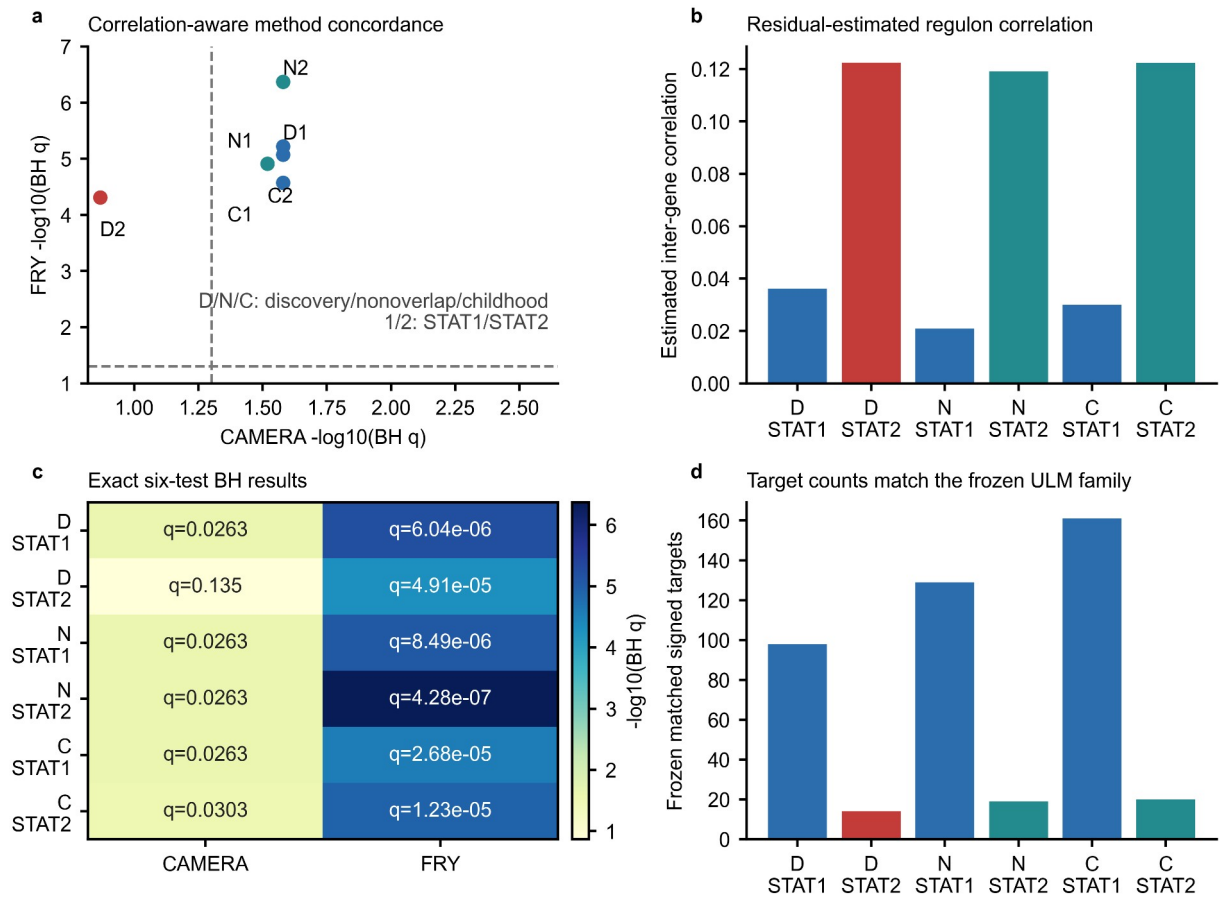
### Supplementary Figure S6 | Independent-validation diagnostics

**a**, Control and SLE donor counts across the five GSE135779 models. **b**, Four-program childhood estimates. **c**, Childhood IFN/ISG estimates after omission of each source B-cell label. **d**, Full childhood program effects and ranges across donor-deletion analyses.



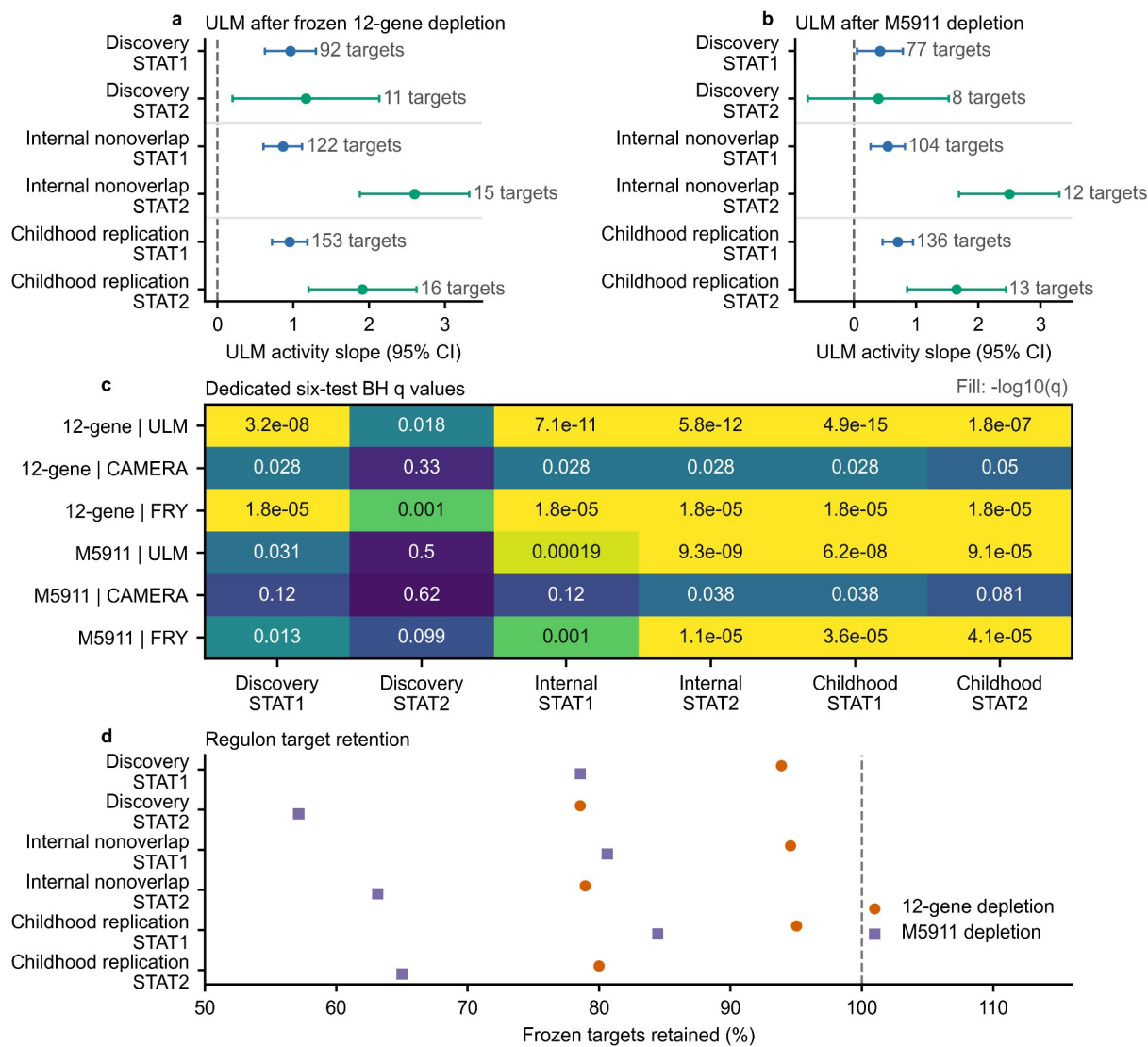
Supplementary Figure S7 | Correlation-aware regulator sensitivity

**a**, CAMERA and FRY six-test BH concordance for STAT1 and STAT2. Dashed guides indicate  $q=0.05$ . **b**, CAMERA residual-estimated inter-gene correlations. **c**, Exact CAMERA and FRY BH  $q$  values. **d**, Matched signed-target counts, which equal the frozen ULM family (STAT1: 98, 129 and 161; STAT2: 14, 19 and 20).



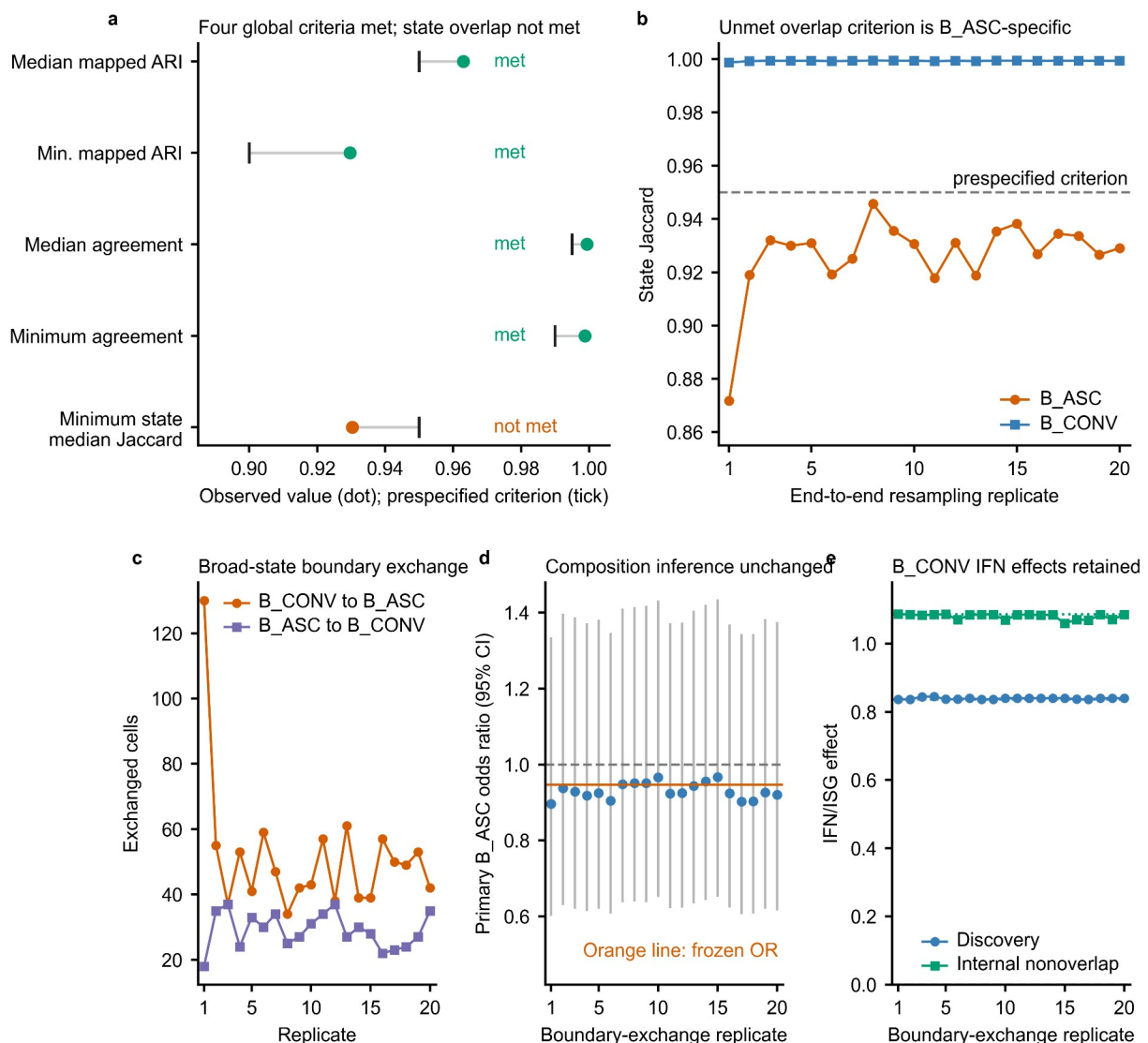
Supplementary Figure S8 | STAT1/STAT2 IFN-overlap-depletion sensitivity

**a**, ULM slopes and 95% confidence intervals after removal of the frozen 12-gene IFN/ISG positive arm. **b**, Corresponding estimates after removal of all 97 M5911 genes; discovery STAT2 retains eight targets and its interval crosses zero. Labels report remaining matched targets. **c**, Exact branch- and method-specific Benjamini-Hochberg q values across the six regulator-by-contrast tests. **d**, Percentage of frozen matched targets retained after each depletion. All method-level directions remain positive, but M5911 depletion produces substantial attenuation and does not support an overlap-independent interpretation.



## Supplementary Figure S9 | End-to-end reconstruction exposes a B\_ASC-specific boundary without changing the disease conclusions

**a**, Observed values and prespecified criteria for five end-to-end two-compartment checks; four met their criteria and minimum state-median Jaccard did not. **b**, State Jaccard across 20 complete reconstruction replicates localizes the unmet overlap criterion to B\_ASC, while B\_CONV remains above 0.95. **c**, Counts of sampled cells exchanged across the B\_CONV/B\_ASC boundary. **d**, Primary B\_ASC composition odds ratios and 95% confidence intervals after each observed boundary exchange; the dashed guide marks one and the orange line marks the frozen estimate. **e**, Primary and donor-nonoverlap B\_CONV IFN/ISG effects after boundary-cell raw counts were propagated through frozen TMM logCPM and HC3 models; dotted lines mark the frozen effects. All propagation analyses reuse GSE174188 and quantify assignment sensitivity rather than independent replication.



## Supplementary Figure S10 | Reference calibration limits source-label-independent external transfer

**a**, Median and interquartile range of full-library divided by selected-feature counts among 13,000 B\_CONV and 1,300 B\_ASC reference training cells. This ratio quantifies the legacy pre-log scaling discrepancy; corrected mapping uses full-library denominators in both datasets. **b**, Per-state precision at the diagnostic elastic-net and eligible centroid thresholds. Mapper colour is held constant across panels b-d, marker shape distinguishes B\_CONV from B\_ASC, and the dashed line is the unchanged 0.90 state-precision criterion. **c**, Reference-cell coverage at those thresholds; the dashed line is the unchanged 0.80 criterion. **d**, Balanced accuracy in each of five donor-grouped reference calibration folds; black horizontal bars show fold means. No eligibility guide is drawn because balanced accuracy is diagnostic only. These folds select model and threshold parameters, not independent performance estimates. The elastic-net B\_ASC precision failure prevents outcome access; centroid success is not a replacement analysis. No corrected external disease result is shown.

